Date ___/___/___

Page _____

Assignment -3

Section-A

Ans-1 Clocking methodology is a synchronization mechanism in processors where all operations are coordinated using a clock signal. Each clock pulse triggers data movement between registers and executions in functional units, ensuring orderly & timed execution of instructions.

Ans 2 A clock cycle is the smallest unit of time in processor operation. Each instruction is broken into steps executed in one or more clock cycles.

Ans-3 Amdahl's law → defines the theoretical speed up of system when a part of it is improved.

$$\text{Speed up} \rightarrow \frac{1}{(1-p)+(p/s)}$$

p = parallel portion of program
s = speed up of that portion

Ans-4 The datapath consists of hardware components responsible for processing data such as ALU, register, buses and multiplexus.

WORLDONE

Date ___/___/___

Page _____

Ans→5 The control unit directs the operation of the processor by generating control signals. It determines which operation the ALU performs, when data is moved b/w components & which instruction is executed.

Ans→6 A single cycle processor executes an entire instruction in one clock cycle. It has a single design but requires a long clock cycle, making it slower & less efficient.

Ans→7 A multi-cycle processor breaks instruction execution into multiple steps. Each step is executed in separate clock cycle. It has a shorter clock cycle time & better hardware utilization compared to single cycle processor.

Ans→8 Instruction pipelining is a technique in which execution of multiple instruction is overlapped by dividing instruction cycle into a number of stages.

Ans→9 ILP is ability of a processor to execute simultaneously by exploiting independent instructions in programm.

WORLDONE

Date ___/___/___

Page _____

Ans-10 Data hazards occurs in pipelining when an instruction depends on the result of a previous instructions which has not completed its execution

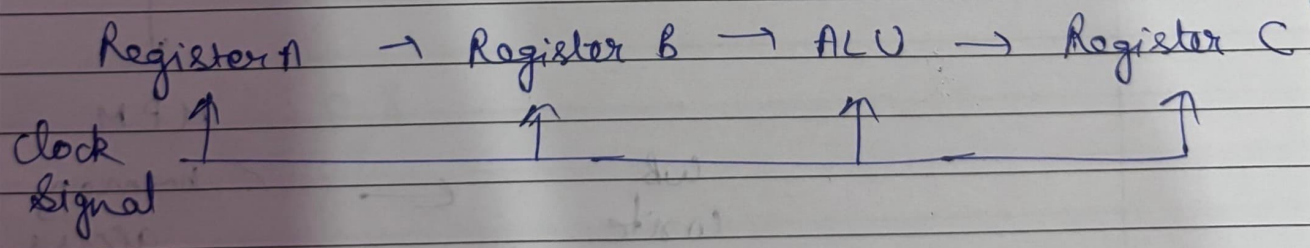
Ans-11 Control hazards occurs due to branch instructions which change the normal flow of program. The pipeline doesn't know which instruction is fetch next.

Ans-12 Pipeline hazards can be mitigated using

- Forwarding (data forwarding)
- Pipeline stalls (inserting bubbles)
- Branch prediction
- Instruction re-ordering (compiller level)

Section - B

Ans-1



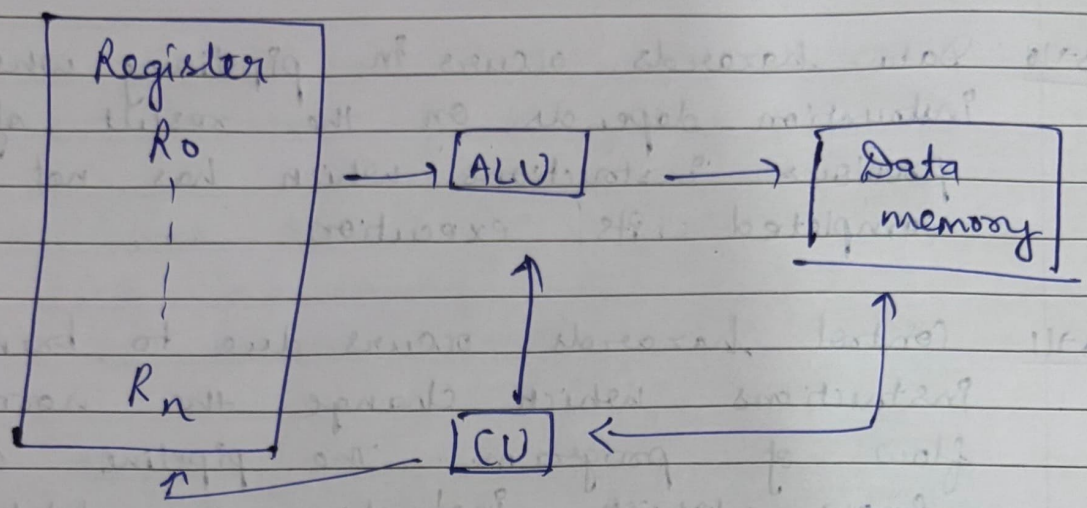
The clock signal synchronizes the operations of all registers and functional units. On each clock pulse, data is transferred and operations are performed

WORLDONE

Date ___/___/___

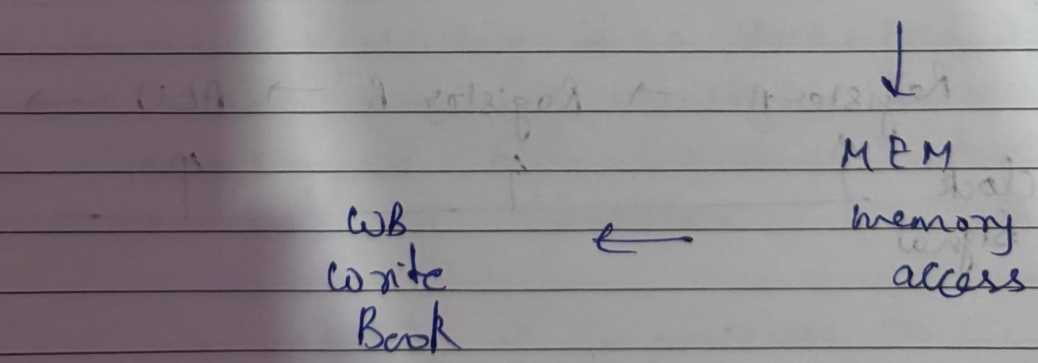
Page _____

Ans→2



Data flows from register to ALU for processing, then to memory or back to register. CU generates signals to control the whole operation.

Ans→3.



WORLDONE

Date ___/___/___

Page _____

Ans→4

(a) Single cycle processor

IF	ID	EX	MEM	WB

Entire instruction executed in 1 long clock cycle.

(b) Multicycle processor

IF	ID	EX	MEM	WB

each stage is executed in different clock cycle.

Section - C

Ans-1 Data hazards will occur. They can be solved using data forwarding (bypassing) or pipeline stalls (inserting bubbles)

Ans→2 $P = 0.8$

$S = 4$

$$\text{Speed up} = \frac{1}{(1-P)+(P/S)} \Rightarrow \frac{1}{0.2+0.2} = \frac{1}{5} \text{ or } 0.2$$

WORLDONE

Date ___/___/___

Page _____

Ans→3 Multicycle processor has shorter clock time and reuses hardware components for different stages. Hence, it is more efficient and has better hardware utilization.

Ans→4 Control hazards occurs. It can be reduced by using branch prediction techniques ie instruction pre-fetching.

Section - D

Ans→1

IF → ID → EX → MEM → WB

Ex stage → In this stage, ALU performs arithmetic or logical operations on the operands. This stage is critical for data processing.

WORLDONE

Date __/__/__
Page _____

Ans→3 Example

ADD	R_1	R_2	R_3	$R_1 = R_2 + R_3$
SUB	R_4	R_1	R_5	$R_4 = R_1 - R_5$

prob = SUB needs R_1 before ADD write its back

Sol^n → use data forwarding from EX / MEM stage to EX stage or insert one stall.

Section - 6

Ans→1 2 assumptions while explaining pipelins

→ it inc. instruction throughput by overlapping

→ execution of multiple instructions

Ans→2 Pipeline hazards may cause stalls, reducing the performance improvement

Ans→3 Pipelining is most challenging because it involves multiple stages hazards & technique to handle them, making it complex to visualize and implement.

WORLDONE